EXPENSE TRACKER – PROJECT REPORT

Project Title: Simple Expense Tracker

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Course: Fundamentals in AIML

Introduction

Managing daily expenses is an essential activity for students, professionals, and households. Small, untracked expenses often accumulate and lead to poor budgeting decisions. The Simple Expense Tracker Program is a Python-based console application designed to help users record, view, and analyze their expenses. It provides category-wise breakdowns, visualization using pie charts, and a minimal, user-friendly workflow suitable for beginners.

Problem Statement

Users struggle to keep track of their daily spending, which leads to ineffective budgeting and poor financial awareness. There is a need for a simple and easy-to-use software application that:

- Records expenses with names, amounts, and categories
- Displays all expenses clearly
- Generates summary reports and visual insights
- Helps users identify their highest spending category

This project addresses these needs by implementing a basic but functional expense tracking tool.

Functional Requirements

Core Functional Modules

1. **Add Expense**
   - Input: Expense name, amount, category
   - Output: Confirmation and storage of the expense
2. **View All Expenses**
   - Display all stored expenses with index, name, amount, and category
3. **Generate Expense Report**
   - Calculate total spending
   - Compute category-wise totals
   - Identify the category with the highest spending

- Display a pie chart summarizing the expenses

Input/Output Structure

Function	Input	Output
Add Expense	name, amount, category	expense entry saved
View Expenses	—	list of expenses
Expense Report	—	totals + analysis + pie chart

Workflow Overview

- User launches the program
- Selects an option from the menu
- Performs add/view/report actions
- Program continues until user chooses Exit

Non-Functional Requirements

Performance

- The system should capture inputs instantly and generate reports with minimal delay.

Security

- No external storage; data stored temporarily in memory during program execution.

Usability

- Menu-driven approach for simple navigation.
- Clear prompts and error messages.

Reliability

- Handles invalid numeric input safely using try/except.

Scalability

- Can be extended to support databases, more categories, multi-user access, etc.

Maintainability

- Modular functions make it easy to update and modify.

Error Handling Strategy

- Validation for numeric input

- Default category assignment for invalid selections
- Safe handling of empty expense lists

System Architecture

The system follows a **simple procedural architecture**:

```
User Input → Menu Logic → Function Calls → Expense List → Report/Visualization
```

Core components:

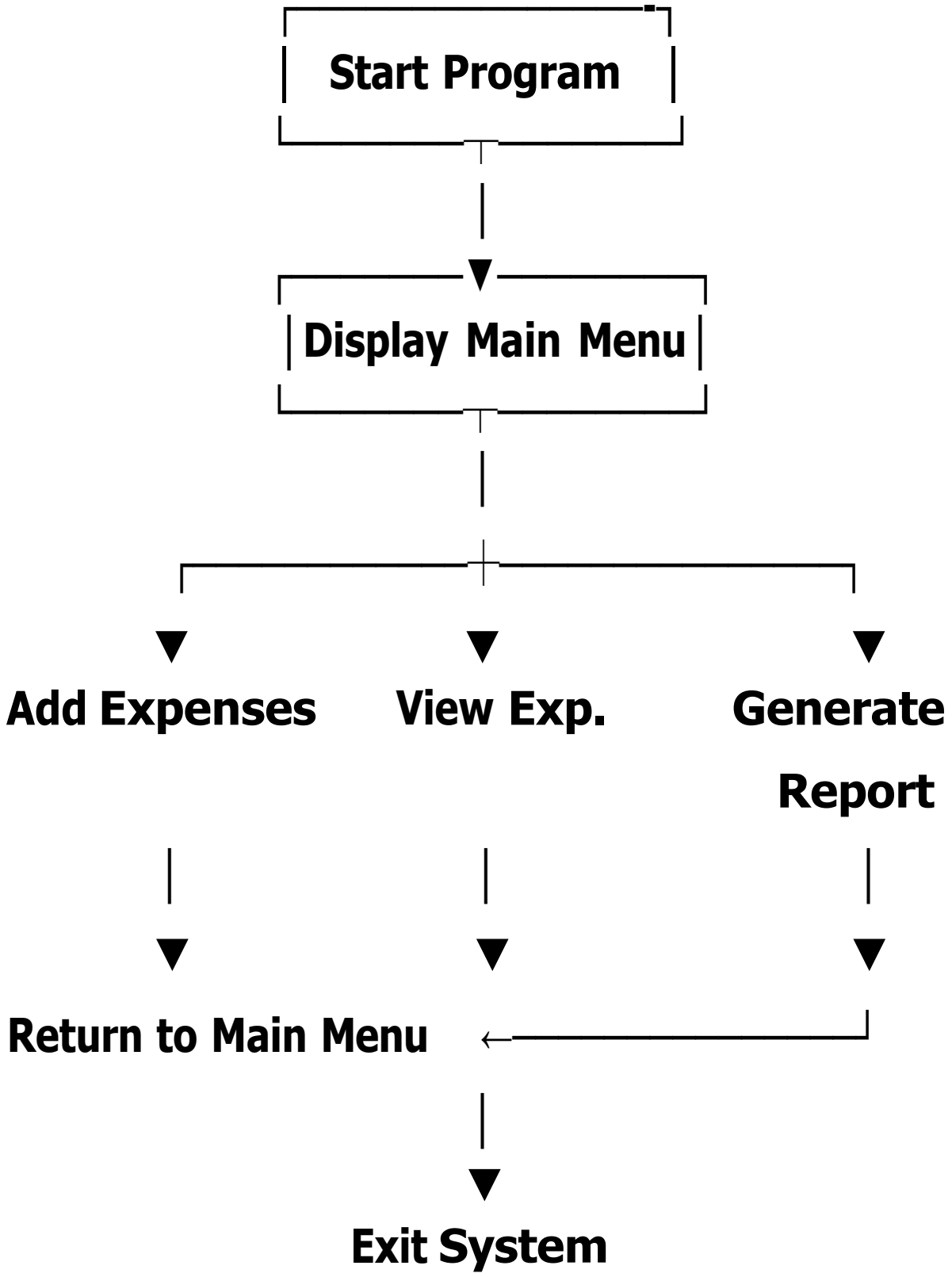
- UI Layer (Console Menu)
- Processing Layer (Functions for operations)
- Data Layer (List storing expenses)
- Visualization Layer (Matplotlib Chart)

Design Diagrams

a) Case Diagram

```
+-----------+
|   User    |
+-----+-----+
      |
interacts with
      V
+---------------------+
| Expense Tracker System |
+---------------------+
| • Add Expense       |
| • View Expenses     |
| • Generate Report   |
| • Exit System       |
+---------------------+
```

b) Workflow Diagram



c) Sequence Diagram

~ User → Main Menu : Select option

~ Main Menu → Add Expense : If option = 1

~ Add Expense → User : Ask for name, amount, category

~ Add Expense → Expense List : Store entry

~ Main Menu → View Expense : If option = 2

~ View Expense → User : Display all expenses

~ Main Menu → Report : If option = 3

~ Report → User : Display totals C chart

~ User → Main Menu : Repeat or exit

d) Class / Component Diagram

(Although program uses procedural style, logical components are shown)

```
+- - - - - - - - - - - - - - - - - - - +
|      Expense App                    |
+- - - - - - - - - - - - - - - - - - - +
| - expenses: list                    |
+-----------------------------------+
| + add_expense( )                    |
| + view_expenses()                   |
| + expense_report()                  |
| + main()                            |
+- - - - - - - - - - - - - - - - - - - +
```

Design Decisions s Rationale

- **Lists used for data storage** because they are simple and require no database setup.
- **Console-based UI** chosen for ease of implementation and accessibility.
- **Matplotlib for visualization** to provide instant visual insights.
- **Module-based functions** for maintainability and readability.
- **Category validation** ensures consistent reporting.

Implementation Details

- Language: Python 3
- Libraries used:

  ~ matplotlib.pyplot for charts
- Program flow:

  ~Main menu triggers functions based on user input

  ~Data stored temporarily in a list of dictionaries

  ~Report function calculates totals and generates pie chart

Screenshots / Results

```
Code  Share  Notebook  Python [conda env:base] *

==================================
          EXPENSE TRACKER
==================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
==================================
Enter your choice (1-4):  1

--- Add New Expense ---
Enter expense name:  dosa
Enter amount:  120
Enter category (Food/Travel/Shopping/Other):  food
Expense added successfully!

==================================
          EXPENSE TRACKER
==================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
==================================
Enter your choice (1-4):  1

--- Add New Expense ---
Enter expense name:  butter paneer masala
Enter amount:  280
Enter category (Food/Travel/Shopping/Other):  food
Expense added successfully!

==================================
          EXPENSE TRACKER
==================================
env:base] * | Idle
```

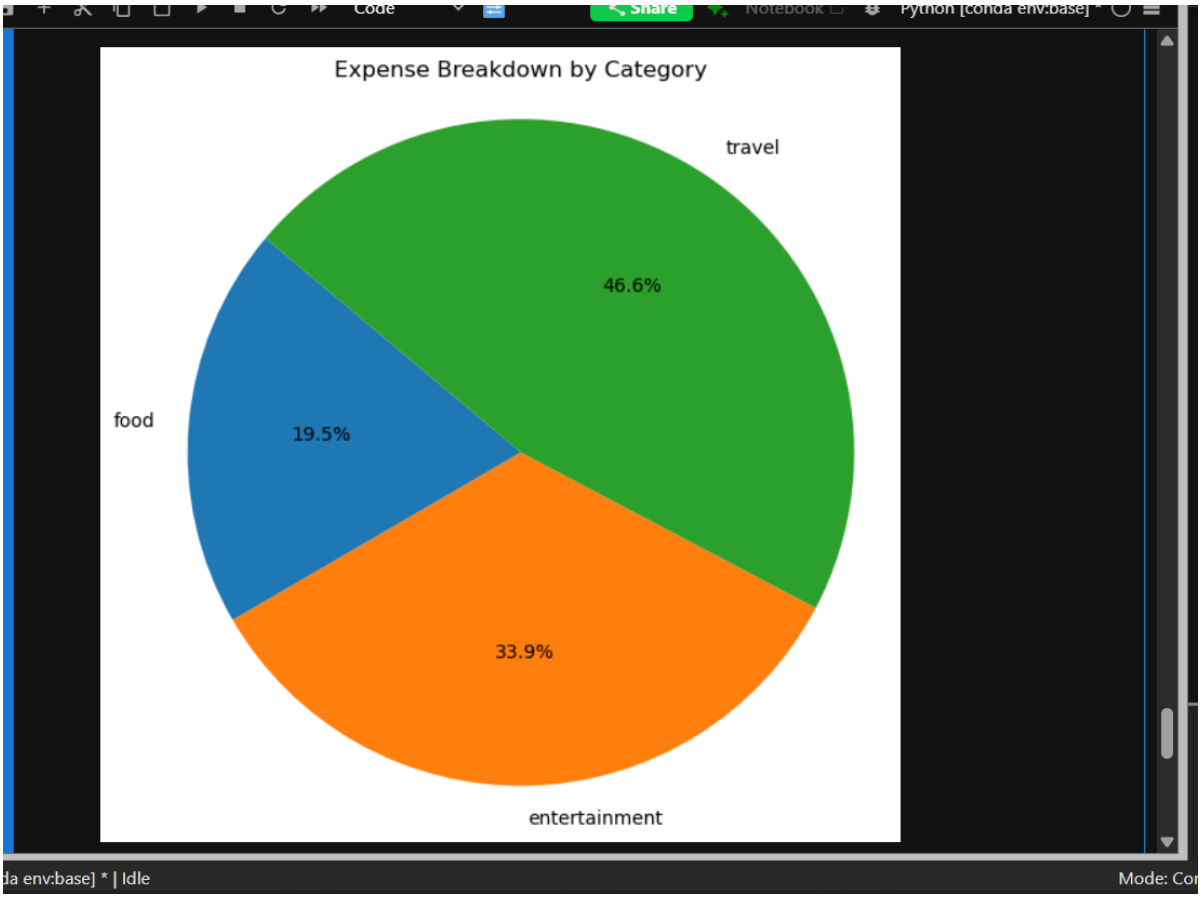
```
          EXPENSE TRACKER
==================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
==================================
Enter your choice (1-4):  butter naan
Invalid choice. Please try again.

==================================
          EXPENSE TRACKER
==================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
==================================
Enter your choice (1-4):  60
Invalid choice. Please try again.

==================================
          EXPENSE TRACKER
==================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
==================================
Enter your choice (1-4):  1

--- Add New Expense ---
Enter expense name:  butter naan
Enter amount:  60
```

```
--- Add New Expense ---
Enter expense name:  butter naan
Enter amount:  60
Enter category (Food/Travel/Shopping/Other):  food
Expense added successfully!

=====================================
          EXPENSE TRACKER
=====================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
=====================================
Enter your choice (1-4):  1

--- Add New Expense ---
Enter expense name:  movies
Enter amount:  600
Enter category (Food/Travel/Shopping/Other):  entertainment
Expense added successfully!

=====================================
          EXPENSE TRACKER
=====================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
=====================================
Enter your choice (1-4):  1

--- Add New Expense ---
Enter expense name:  games
Enter amount:  200
```

```
--- Add New Expense ---
Enter expense name:  games
Enter amount:  200
Enter category (Food/Travel/Shopping/Other):  entertainment
Expense added successfully!

=====================================
          EXPENSE TRACKER
=====================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
=====================================
Enter your choice (1-4):  1

--- Add New Expense ---
Enter expense name:  bus
Enter amount:  100
Enter category (Food/Travel/Shopping/Other):  travel
Expense added successfully!

=====================================
          EXPENSE TRACKER
=====================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
=====================================
Enter your choice (1-4):  1

--- Add New Expense ---
Enter expense name:  train
Enter amount:  1000
```

```
--- Add New Expense ---
Enter expense name:  train
Enter amount:  1000
Enter category (Food/Travel/Shopping/Other):  travel
Expense added successfully!

====================================
          EXPENSE TRACKER
====================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
====================================
Enter your choice (1-4):  2

===== ALL EXPENSES =====
1. dosa            | ₹ 120.00 | Category: food
2. butter paneer masala | ₹ 280.00 | Category: food
3. butter naan     | ₹  60.00 | Category: food
4. movies          | ₹ 600.00 | Category: entertainment
5. games           | ₹ 200.00 | Category: entertainment
6. bus             | ₹ 100.00 | Category: travel
7. train           | ₹1000.00 | Category: travel
------------------------------

====================================
          EXPENSE TRACKER
====================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
====================================
Enter your choice (1-4):  3
```

```
Enter your choice (1-4):  3

===== EXPENSE REPORT =====
Total Expenses: ₹2360.00

Category-wise Breakdown (With Percentages):
- food        : ₹460.00 (19.49%)
- entertainment: ₹800.00 (33.90%)
- travel      : ₹1100.00 (46.61%)

===== FEEDBACK =====
ℹ️ 46.6% of total expenses went to travel. Watch this category.
------------------------------
```




```
=====================================
          EXPENSE TRACKER
=====================================
1. Add Expense
2. View All Expenses
3. Generate Expense Report
4. Exit
=====================================
Enter your choice (1-4):  4
Exiting program... Goodbye!
```

Testing Approach

Manual Testing

- Tested input validation with:
  - Correct numeric values
  - Incorrect values (text, empty input, symbols)
- Tested category selection
- Tested empty list behavior for:
  - View expenses
  - Report generation

Functional Testing

- Verified all menu options produce correct output.

Boundary Testing

- Very large amount values

- Repeated entries
- Only one category type

Challenges Faced

- Ensuring robust input validation for numeric values
- Implementing category-wise calculations manually
- Handling cases with no expenses entered
- Integrating matplotlib pie chart in a console application

- -

Learnings s Key Takeaways

- Gained understanding of modular programming in Python
- Improved ability to validate and sanitize user input
- Learned how to generate charts with Matplotlib
- Understood how to structure simple software requirements and architecture
- Experienced designing user-friendly console workflows

Future Enhancements

- Add permanent storage (JSON, CSV, or database)
- Add authentication for different users
- Add monthly/yearly reports
- Export reports to PDF
- Build a GUI version using Tkinter or PyQt
- Add advanced categories and analytics

References

- Python official documentation: https://docs.python.org/
- Matplotlib documentation: https://matplotlib.org/
- Online Python tutorials and learning resources
- Class lecture notes and assignments